Conditions of Trust: What an AI Must Grasp Before We Rely on It

Abstract

When a doctor, a judge, or a teacher must decide whether to defer to an AI model's output, they cannot avoid asking what the model has actually grasped—yet our inherited concept of understanding, freighted with phenomenological and normative commitments that fit only human minds, is ill-suited to answering them. This paper argues that the cognitive register is nonetheless indispensable: any vocabulary fine-grained enough to separate trustworthy outputs from lucky ones inevitably reintroduces the inferential core of *understanding*, so the task is not to abandon the concept but to re-engineer it for the AI case—preserving its tie to structure-tracking competence, severing its human-specific implications, and forging new links to the internal structure that recent interpretability research can expose. Because that structure is *polyphonic*, spread across many parallel and uneven mechanisms rather than lodged in a single locus, attributions of understanding must be task-indexed, graded, and defeasible, and the trust they license is something a model's inspectable organisation earns rather than something its fluency presumes. The upshot is a sober framework of cumulative, calibrated indicators—mechanistic and behavioural—for judging, condition by condition, how far an AI's grasp of a domain warrants our reliance.

1. The question of understanding: a three-way split

The question whether AI models *understand* anything has provoked sharply diverging responses. One group liberally anthropomorphises AI, applying existing cognitive concepts such as *understanding*, *reasoning*, and *intelligence* to these systems (Piantadosi and Hill 2022; Bubeck et al. 2023; Søgaard 2023; Mandelkern and Linzen 2024; Brooks et al. 2024; Cappelen and Dever 2021). A second group categorically denies the applicability of cognitive concepts to such models and reduces their achievements to superficial statistics, conceptualising them as "stochastic parrots" (Bender, Gebru, McMillan-Major & Shmitchell 2021), "imitation engines" (Yiu, Kosoy & Gopnik 2023), or "pattern matchers" (Weber 2024). A third group dismisses the debate as a futile distraction: as long as the models produce the outputs we want, we should not worry too much about how they got there (Krishnan 2020; Carlini & Scarfe 2025). On this last view, it is a merely "philosophical" question in the pejorative sense of the term—an idle speculation that has hardened into a persistent and polarised debate only because one's answer depends on the *conception* of understanding one implicitly presupposes. Anthropomorphisers rely on conceptions of understanding that accommodate AI systems, while statistical reductionists rely on conceptions that exclude them. That tells us less about AI systems than about the conceptual predilections of each party to the debate.

We propose to break this impasse by re-engineering the concept of understanding for the AI context. Our approach is metaconceptual and functionalist: instead of arguing whether AI fits our ordinary concept, we ask what *function* that concept performs in human affairs, and how a successor concept might serve the same function in our dealings with AI systems. Against the indifferentists, we argue that the question whether AI "understands" is anything but idle: it needs to be asked, and a great deal turns on the answer. It is true that one's answer depends on one's conception of understanding; but the conclusion to draw from this is not that the question is empty, but that the debate must ascend to the metaconceptual level and confront the question of *how best to conceptualise* understanding in the context of AI systems. Against the reductionists, we maintain that something like the concept of understanding is required to discriminate *between* AI models, and between what a given model *can* and *cannot* do. Against the anthropomorphisers, we note that our existing concept carries human-specific implications—phenomenological and normative—that do not transfer to AI. We therefore need to adapt the concept and engineer a successor notion—*machine understanding*—that can be rigorously applied to AI models.

We defend a three-part thesis. First, the cognitive register is not optional: the deployment of some conception of understanding is *indispensable* to human–AI interaction. The master argument here is what we call the *reintroduction argument*: any vocabulary capable of doing the discriminative work that human–AI interaction requires of us is bound to end up reintroducing the core inferential connections characteristic of the concept of understanding.

Second, inherited conceptions of understanding, taken as they stand, are *inadequate*, because they carry commitments and implications—phenomenological, doxastic, and normative—that do not fit AI systems.

Third, the task is therefore to *adapt* the concept of understanding to fit AI through three operations: *preserving* the inferential connections that do real work in transferring to the AI case; *severing* the connections that hold only in the human case and mislead when applied to AI; and *forging* new connections to the mechanistic indicators offered up by recent interpretability research. The product is a conception of *machine understanding* that is responsive to our conceptual needs and operationalisable through a cumulative-warrant framework of mechanistic indicators vindicated by interpretability evidence.

The paper proceeds as follows. Section 2 makes the case that the cognitive register is indispensable, mounting two arguments—the *insufficiency argument* and the *reintroduction argument*—against the deflationist who would describe AI in purely statistical terms. Section 3 turns to the adaptation itself, identifying the kinds of internal structure—features, connections between features, and content-independent circuits—onto which an attribution of machine understanding can be forged, and showing through two interpretability cases how such structure can license trust. Section 4 confronts the first complication, *polyphony*: a model's competence is typically spread across many parallel and uneven mechanisms rather than lodged in a single circuit—a picture we render intuitive through the analogy of a jury room, and whose consequences for attributing understanding we then draw out. Section 5 confronts the second complication, scale: the frontier models we most need to judge are too large to reverse-engineer in full, yet we argue that scale tilts the prior towards structure rather than memorisation, and that a graded, defeasible warrant can be assembled from mechanistic and behavioural indicators calibrated against the cases we can fully verify. Section 6 returns to the contrast between a unified human understander and a polyphonic model, arguing that the coherence the concept presupposes is, in both cases, achieved through elicitation rather than intrinsic to the substrate. A brief conclusion takes stock.

2. Why the concept of understanding is indispensable

The three-way split sketched in §1 might make the question of how to conceptualise AI cognition look academic—a verbal dispute between camps with different definitional preferences. It is not. In a growing range of practical situations, agents must decide whether and how to defer to AI outputs, and they cannot do so without taking a stand on what the model has and has not grasped. The indifferentist's dismissal of the debate as idle speculation is sustainable only if one looks away from the practical situations in which the question of understanding becomes acute.

Consider a doctor consulting an AI model about a perplexing case. Suppose the model proposes an intriguing differential diagnosis. The doctor must now decide how far to trust it. But for this she needs to know whether the model has *understood* the disease—whether it has grasped the causal structure connecting causes, symptoms, and treatments—or whether it is merely producing a string of plausible-sounding medical text on the basis of superficial correlations (text including the medical terms "X" and "Y", say, tends to be followed by text including the term "Z"). The concept of understanding does real practical work here: it guides the allocation of epistemic trust.

Nor is the doctor's predicament unique. A judge weighing AI-generated sentencing recommendations must likewise discriminate between recommendations grounded in a grasp of how the pertinent legal precedents bear on the present case and recommendations that pattern-match on the surface features of comparable cases; a professor deciding whether to encourage students to use AI tutors for a difficult topic must determine whether the model's command of the material is deep enough to warrant her endorsement. These practitioners cannot shirk the discriminative task by dismissing the understanding question as philosophical; they have a real *conceptual need*—a need for a concept to do the work traditionally performed by the concept of understanding.

This is why indifferentism is unsustainable as a response to the understanding question. The question *demands* an answer. And the point is not that *we* as theorists need to address it; it is that *practitioners*—doctors, judges, and teachers—need to. The question is ineluctable in practice.

For theorists, this in turn means that the question cannot simply remain whether the inherited concept of understanding applies *to AI systems* as a class, or to some specific class of them. Since the conceptual work to be done is to differentiate not just between AI systems, but between reliable and unreliable outputs in concrete cases, the question at the level of theory must be how our conceptual repertoire should be configured to discriminate among AI capacities for the purpose of allocating trust and calibrating expectations. This reframing is itself a theoretical contribution: it transmutes a stalemated dispute into a research programme for conceptual engineering.

The deflationist about AI cognition does not shirk the question of understanding, but denies both that the concept applies and that cognitive vocabulary is required to navigate human–AI interactions responsibly. Instead of speaking of these systems as exhibiting "understanding", "comprehension", or a "grasp" of things, the deflationist holds, we should describe them as performing "matrix multiplications" that produce outputs entirely on the basis of "superficial correlations", "distributional statistics", or "pattern-matching". Bender, Gebru, McMillan-Major, and Shmitchell (2021) have made the case under the banner of "stochastic parrots"; Marcus (2018), Floridi (2023), Bishop (2021), and Bender and Hanna (2025) have made it under various others ("mathy maths", "text-extruding machines", "conversation simulators").

Our contention is that this deflationist strategy cannot be sustained. We offer two arguments for this conclusion: the *insufficiency argument* and the *reintroduction argument*.

2.1 The insufficiency argument

One can, of course, describe what AI models do without reaching for cognitive vocabulary. At the level of the implemented computation, the deflationary description is perfectly apt: a transformer converts its input into numbers, or “embeddings”, passes those embeddings through layers of attention mechanisms, matrix multiplications, normalisations, and non-linear transformations, and returns a probability distribution over its vocabulary of tokens, from which the next token is then picked according to simple rules. There is no hidden homunculus inside the model—it is tensor operations all the way down. In that sense the deflationist is right to insist that the system is a mathematical object: a very large function taking words as input and returning a probability distribution over all possible next tokens as output.

The question is what follows from this. Our answer is: much less than the deflationist needs. The mathematical description is perfectly correct, but too indiscriminate. It applies equally when the model gives a good medical diagnosis and when it fabricates one; when it solves an arithmetic problem by a robust procedure and when it lands on the right answer by a brittle heuristic. In each case the output is produced by the same broad kind of operation. To say that it was produced by matrix multiplications is therefore not yet to say anything that helps practitioners decide whether the output should be trusted.

This insufficiency is not an objection to the mathematical description *per se*. It is an objection to mistaking an *implementation-level* description for a *capacity-level* one. Even a complete mathematical specification of a forward pass through the model would not, by itself, tell us what abstractions the model is using, whether those abstractions are reliable, whether the output depends on a shortcut, or how far the procedure will generalise beyond the cases on which the model was trained or tested. Yet these are precisely the questions that matter practically—the questions we address when we ask what exactly the model has “understood”.

The point should be especially clear to those who are versed in the mathematics of these systems. Precisely because they know that the model is a stack of tensor operations, they also know that this description remains invariant across the very cases they most need to distinguish. It tells them that the sound diagnosis and the hallucination were both produced by a forward pass through the network; it does not supply the conceptual resources needed to tell the two apart.

The implication is that, given the place these models increasingly hold in our lives, we need to conceptualise them at higher levels of description in order to bring into view the distinctions that matter to us. Just as an exhaustive description of the physics of the human body is silent on whether the body is healthy, so a mathematical description of the computational operations performed by an AI model is silent on whether the model is merely matching surface patterns or drawing on joint-carving abstractions formed during training; whether it is relying on shallow heuristics or on robust connections that actually track the structure of the domain; whether it is myopically producing one token at a time, with no anticipation of what follows, or planning several tokens ahead to ensure consistency and coherence at a larger scale. The mathematical description is *insufficient* because it does not supply the discriminations required for the allocation of trust and the calibration of expectations.

2.2 The reintroduction argument

Any vocabulary capable of doing the discriminative work that human–AI interaction requires of us will inevitably reproduce some of the core inferential patterns that characterise the concept of understanding. Suppose we go along with the deflationist's proposal and try to dispense entirely with cognitive vocabulary in thinking and talking about AI systems. Can this be sustained without covertly reintroducing cognitive concepts such as *understanding*?

Return to the doctor. Even if she takes the deflationist's advice and tries to think about her predicament in purely non-cognitive terms—purely in terms of pattern-matching, say—she cannot avoid the practical question of whether to trust the model's output. And this practical question forces a discrimination between different kinds of pattern-matching. Not all pattern-matching is equal: some kinds merit deference, others do not, and the doctor needs to tell them apart. But what *is* the kind of pattern-matching that merits deference? It is pattern-matching that tracks the underlying structure of the domain—that is sensitive to the causal dependencies between causes, symptoms, and treatments, rather than to the superficial correlations between them. And that just redescribes the conditions under which we apply the concept of *understanding*, and links them to epistemic trustworthiness. The inferential connections drawn by our cognitive concepts are being reintroduced through the back door.

This is what we call the *reintroduction argument*: any attempt to draw the distinction between reliable and unreliable AI outputs in deflationary vocabulary will reintroduce, under whatever label, some of the very inferential connections that vocabulary sought to abjure. The fundamental reason is that non-cognitive vocabulary is *expressively inadequate* to the task of discriminating between the finer shades of competence and reliability that we must distinguish in dealing with systems as sophisticated as contemporary AI models.

To see this, consider what happens if our doctor recognises the practical need for finer conceptual discrimination but remains resolved to avoid cognitive vocabulary. She might seek the necessary resources in more sophisticated statistical notions that help her qualify the kind of "pattern-matching" she is dealing with. We canvass three such deflationary strategies, in increasing order of sophistication, and show that each either fails or covertly reintroduces what it was meant to replace.

The first and most obvious way to discriminate in purely statistical terms between more and less reliable kinds of pattern-matching is by appeal to *track records of accuracy*. Such track records are readily available in the form of benchmark-performance statistics. If the model scored 95% on a medical-diagnosis benchmark (such as CPC-Bench, MSDiagnosis, or the Sequential Diagnosis Benchmark), this would be evidence of its reliability and would suggest that the doctor should defer to it.

But, as the deflationary literature itself stresses, this criterion is insufficient. Models can achieve impressive scores on benchmarks and still fail on real-world examples. High benchmark performance is compatible with shortcut learning: the model may have learned to rely on superficial cues that happen to predict the right answer in the benchmark cases but are not actually tied to the phenomenon at issue. Such cues may carry the model through the test—especially if the test set is drawn from the same distribution as the training set—while offering little assurance that the model will be accurate once confronted with a different real-world case. Nikankin et al. (2025) show that frontier models perform multi-digit arithmetic via a bag of heuristics that are effective within the training and testing setting but break down outside it; their reliability is not zero, and yet the models' "grasp" of arithmetic—to use the question-begging term we are arguing toward—is plainly partial. Raw records of accuracy thus cannot deliver the discrimination the doctor needs, because they fail to distinguish a trustworthy model scoring 95% on a benchmark from an untrustworthy one with the same score.

A second strategy, which promises to remedy the shortcomings of the first, would be to look to a model's real-world track record of reliability across similar cases. This acknowledges that real-world reliability is the relevant property. But even if the relevant data were available, what counts as a relevantly similar case? The doctor needs a verdict about *this* case, here and now; and it was precisely because the case was not run-of-the-mill, but in crucial respects unique, that she turned to the AI model in the first place. A track record of reliability on loosely similar cases tells her little about whether the present output is reliable. And *which* cases are relevantly similar ultimately depends on what the particular causal structure of the particular disease at work in the present case *is*—which is exactly what she is trying to find out by asking the model, and so cannot help her decide whether to trust its answer.

Third, the doctor can try to discriminate between more and less reliable pattern-matching by appealing to causal-interventionist criteria. Following Pearl (2018), the deflationist might say: the model tracks p if and only if its representation of p covaries with interventional dependencies $\mathrm{P(p} \mid \mathrm{do(X = x))}$ rather than merely with observational correlations $\mathrm{P(p} \mid \mathrm{X = x)}$. The notions are to be read as follows. $\mathrm{P(p} \mid \mathrm{X = x)}$ asks how likely p is given that X is *observed* to have the value x; it registers patterns of co-occurrence. $\mathrm{P(p} \mid \mathrm{do(X = x))}$, by contrast, asks how likely p would be if X were actively *set* to x by an intervention while the rest of the system were allowed to respond. The do-operator is meant to distinguish cases in which X merely travels together with p from cases in which changing X would make a difference to p. In the medical case, the first question might ask whether patients with symptom, treatment, or risk factor X also tend to have disease p; the second asks what would happen to p under an intervention on X. If X is a genuine difference-maker, p should change; if X is only a downstream symptom or coincidental marker, the association may disappear once we ask the interventionist question. The interventionist criterion therefore says that a model's internal representation must be sensitive to what would change, and what would remain stable, under relevant interventions in the disease process: a model that is trustworthy in a given case tracks the disease's causal structure, while one that is untrustworthy tracks only co-occurrences. This criterion has the right shape to draw the distinction that needs drawing. Yet it also aligns exactly with one of the criteria for understanding that the philosophical literature has stressed for two decades. Woodward (2003) characterises understanding-why in terms of a grasp of what-would-happen-if relationships—that is, in terms of sensitivity to interventional dependencies; Strevens (2008) treats the kernel of explanatory understanding as the identification of difference-makers in causal structure; Khalifa (2017) systematises the family of accounts under which understanding is constituted by a grasp of modal-structural relations. The deflationist who reaches for causal-interventionist criteria is reintroducing understanding in all but name.

Statistical terms thus either fail to deliver the discrimination the doctor needs—as with track records of accuracy on benchmarks and statistical reliability across similar real-world cases—or deliver it only by reintroducing the inferential structure of cognitive vocabulary under a different label, as with causal-interventionist tracking. This is the reintroduction argument. It brings out that the cognitive register is not optional: some concept doing the discriminative work of *understanding* is indispensable, and whatever concept ends up doing that work will be inferentially continuous with the concept of understanding—that is, it will reproduce the core inferential patterns the concept underwrites. The deflationary alternative is not so much inapplicable as *insufficiently responsive* to what we demand of our concepts, given what human–AI interaction demands of us. We need the cognitive register. Whether the inherited concept itself is fit for the new application, or must be adapted to fit it, is the question of the next section.

3. Circuitry that licenses trust

Some of the internal circuitry of LLMs can license the very kind of trust that a concept of understanding adapted to them should latch onto. Following Beckmann and Queloz (2026), we take the competence of LLMs to be grounded in three different kinds of structure:

- **Features:** directions in latent space that unify the diverse manifestations of an entity or property—such as the "Golden Gate Bridge" feature, which is causally efficacious and steerable.
- **Connections between features:** factual and causal links between features—for example, a "Michael Jordan → basketball" connection that ensures the retrieval of the information that Michael Jordan is a basketball player.
- **Content-independent circuits:** subnetworks that implement a general procedure rather than storing cases. The paradigmatic example is the modular-addition circuit.

These forms of internal circuitry are what an attribution of machine understanding can be forged onto: inspectable structure inside the model that can ground trust. The question, for any given task, is whether the *right kind* of structure is being relied on.

3.1 A medical diagnosis attribution graph

A particularly telling example brings us back to the doctor of §2. Lindsey et al. (2025) put a familiar sort of vignette to Claude 3.5 Haiku—a 32-year-old at 30 weeks' gestation, with severe right-upper-quadrant pain, a mild headache, nausea, a blood pressure of 162/98, and mildly elevated liver enzymes—and ask which single further symptom it would be most useful to enquire about. Its leading completions are visual disturbances and proteinuria. To a clinician these are not arbitrary: both are cardinal signs of preeclampsia, the condition that this constellation of findings should bring to mind.

To see how the model gets there, the authors build an *attribution graph*. A so-called *cross-coder* is trained to disentangle the model's superposed features *for this one inference*, and the result is pruned to the features that carry the computation. What remains is a map of which features activate, and in which order, as the input passes through the layers.

What the graph shows is that the model is not regurgitating a memorised template for the case but relying on the right kind of features and connections, much as a clinician who understands preeclampsia would (Figure 1).

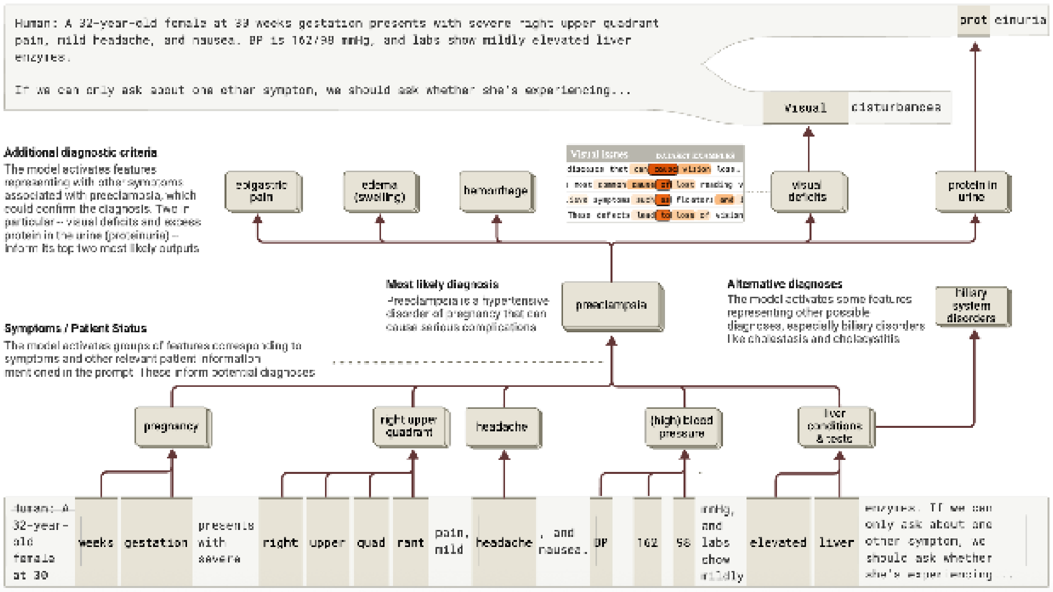

Figure 1. Attribution graph for Claude 3.5 Haiku on the preeclampsia vignette (Lindsey et al. 2025). The patient-status features (*gestation*, *hypertension*, *right-upper-quadrant pain*, *elevated liver enzymes*) activate a feature for *preeclampsia* alongside competing hypotheses (*cholecystitis*, *cholestasis*). The preeclampsia coalition in turn activates downstream features for its confirmatory signs (*visual disturbances*, *proteinuria*), which surface as the model's leading completions.

The model represents the pain, the headache, the high blood pressure, and the raised liver enzymes as related to one another in virtue of their all being compatible with a single diagnosis, preeclampsia. Drawing the scattered symptoms together under one diagnosis in this way is the unification we expect of conceptual understanding (tier 1). The model also keeps rival diagnoses in play, representing cholecystitis and cholestasis more weakly alongside it, as a differential diagnosis should. The connections between features then link the leading diagnosis to the condition's other typical signs (tier 2). Since the question asks which symptom to check next, those signs—visual disturbances and proteinuria—are the ones it gives. So the features are not just present; they are connected in the right way, running from the presenting signs to the diagnosis to the further signs it predicts. That tracks the structure of the condition, not a verbal association.

The *preeclampsia* feature also plays the expected causal role. When the experimenters inhibit the preeclampsia features, the model's recommendation shifts to "decreased appetite", a sign of cholecystitis—the next hypothesis in line. We can see the answer coming from structure that tracks how the underlying domain is actually organised. This warrants a *local* trust, for this particular output in this particular case.

3.2 Addition circuits

A content-independent circuit can warrant more than the local trust of the diagnosis. The clearest case, and one of the first learned circuits to be reverse-engineered in full, comes from a small transformer that Nanda et al. (2023) trained on nothing but modular addition. Modular addition wraps around once it passes a fixed point, the way the hours on a clock run from 12 back to 1. This model was trained on addition modulo 113, so that 55 + 71 comes to 13 rather than 126, the total having gone once around the loop of 113. For a long time it merely memorised the training examples, but then it "grokked" the task and began to generalise to pairs it had never seen.

The discovered procedure is elegant. Rather than add along a line, the circuit represents each number as an angle and sums the angles, so that the running total rotates around a circle. The trick is the frequency—the speed at which it turns: this is set so that one full turn around the circle is exactly the modulus. Once the total passes a full turn, it returns to zero on its own, so the wrap-around of the circle delivers the modular result for free.

The model does not rely on a single circle. It runs the same angle-addition on circles of several frequencies, some turning faster and wrapping around many times. Spreading a quantity across frequencies in this way is a Fourier representation, and the circuit is accordingly known as the Fourier-multiplication algorithm. The several frequencies are not idle redundancy; they pay off through constructive interference. The base circle lands on the right answer, but its peak is broad, so it is accurate without being precise; the faster circles are sharp but also score highly for some wrong answers. Read together, only the genuinely correct answer scores well on all of them, so the model achieves both accuracy and precision (Beckmann & Queloz 2026).

It might be thought that this circle-and-frequency machinery is an artefact of a small network trained on nothing but modular addition. Subsequent work on larger models trained in the ordinary way suggests otherwise. Indeed, such models have been found to compute even *regular* addition with several frequency-based circuits operating at different periods, whose combination yields the same constructive interference, and the robustness that comes with it (Zhou et al. 2024; Nikankin et al. 2025; Kantamneni & Tegmark 2025).

A further question is whether such a circuit operates only when the model is presented with obvious addition in the simplest textual format, such as "2+5". Here too the evidence points the other way. Feucht et al. (2026) show that Llama-3.1-8B relies on a single addition mechanism across a variety of tasks that share an additive structure: not only plain arithmetic, but also questions such as "What month is six months after August?" or "What day is five days after Wednesday?" (Figure 2).

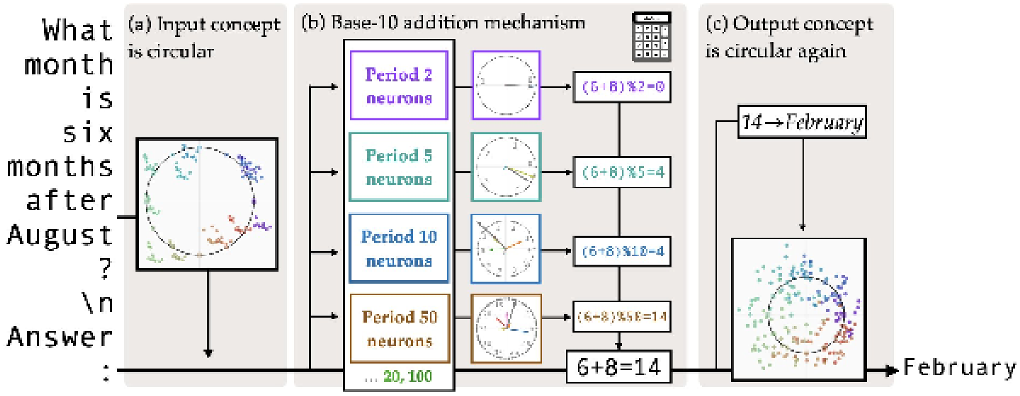

Figure 2. Llama-3.1-8B reuses one base-10 addition mechanism across cyclic concepts (Feucht et al. 2026). For "What month is six months after August?", the model first computes $6 + 8 = 14$ using Fourier-feature neurons at several periods (2, 5, 10, 50, and also 20, 100), then applies the modulus in later layers to map 14 back to a month, February. The same mechanism serves plain arithmetic, weekdays, and clock times.

One might have expected the model to maintain a dedicated modulo-12 calculator for months and a modulo-7 one for weekdays. Instead it relies on the main addition mechanism and only in later layers applies the relevant modulus to return to the appropriate concept (mapping fourteen to February, say). The addition is carried out, once again, on Fourier features at periods 2, 5, 10, 20, 50, and 100. The model thus behaves as though it had grasped a single addition procedure and brought it to bear wherever addition is required—in arithmetic and in cyclic concepts such as months, weekdays, and clock times alike.

What such a circuit licenses is broader than what the diagnosis licensed. There, the attribution graph warranted trust in a single inference. A content-independent circuit gives us reason to trust the model across a whole class of cases at once: implementing the general rule rather than the right output for particular inputs, it can be expected to handle additions it was never shown, not just the one inference we happened to inspect. The work just discussed suggests this scope can be wide. If a single mechanism is reused across bare sums, numbers spelled out in words, months, weekdays, and clock times, then trust in the model's addition extends to every guise addition takes in text. The evidence here comes from one circuit in one model, so it marks a direction rather than a settled generalisation. Still, it shows the kind of broad, class-level trust that a content-independent circuit can underwrite.

Taking the two cases together, we can say what an attribution of machine understanding amounts to. *A model understands a domain* when it has formed sufficiently joint-carving features and linked them into circuits encoding sufficiently general and reliable procedures, so that it can be trusted to produce a correct output with reasonably high probability. The attribution is, at bottom, a claim about internal structure, not about trust: trust is what the structure earns, not what constitutes it. To say that a model understands is to say that, were we to locate the relevant features and circuits and reconstruct the procedure they implement, what we found would vindicate deference—the output would prove to issue from structure-tracking rather than from a shortcut that merely coincides with the structure on the cases encountered so far. The diagnosis and the addition circuit are cases where this condition is met, and where locating the structure and intervening on it shows the trust to be earned rather than lucky. Mechanistic interpretability, in short, can uncover organisation of the very kind that licenses trust.

This is the adaptation §1 promised. We *preserve* the inferential core of the inherited concept—its tie to structure-sensitive competence, to a grasp of how things depend on one another and of what would happen if they were otherwise. We *sever* the human-specific commitments that do not carry over—the felt click of insight, the belief-holding subject, the authority and responsibility of an agent. And we *forge*, in their place, links to the three kinds of internal structure just surveyed. The attribution that results is task-indexed, graded, and defeasible: it licenses epistemic deference proportional to the warrant, without making the model an authority or holding it responsible in the human sense, and it rests on inspectable internal organisation, and interventions upon it, rather than on first-person report or behaviour alone.

Two complications stand between this picture and its everyday use, and they organise the next two sections. The first is that the structure inside a model is seldom a single, isolable circuit. Competence is typically spread across many parallel and partly redundant mechanisms, with no one locus, the output emerging from their interaction. We call this condition *polyphony*, and take it up in §4. The second is that locating and reconstructing such structure is hard and costly, and for the largest models it is largely out of reach. The full mechanistic test can be run on small or simple systems, but not yet, in general, on the frontier models whose outputs we most need to trust. How to license trust under that limitation is the subject of §5.

4. Addressing polyphony

This section takes up the first complication, polyphony. By it we mean that a model's competence is not lodged in one circuit but spread across many mechanisms running in parallel, of uneven reliability, so that any output is the joint product of their interaction rather than the work of a single one. We first make the picture concrete through an analogy, the jury room (§4.1), which also renders a range of interpretability findings intuitive. We then draw out the complications polyphony creates, both for attributing understanding and for trusting any particular output (§4.2).

4.1 The jury room analogy

Imagine a very large group of jurors working together on a long and complex trial. Every morning a new jury gathers in the same room—each day corresponds to one layer of the network, one round of parallel processing. The jurors do not remember previous days; they inherit only the accumulated material on a large communal table. *The table corresponds to an LLM's residual stream.*

Each note on the table is a piece of evidence with a determinate meaning, such as "the defendant was at the scene". *These notes are the analogue of activated features.*

Each day begins with a morning phase. Before doing any substantive work, each juror selects specific pieces of evidence from the table according to a fixed rule or perspective—some look for causal relations, others for numerical regularities, others for fragments resembling past cases, others for surface patterns or keywords. Much of the evidence is ignored, and different jurors often select overlapping pieces. *This selection phase plays the role of attention heads: rule-governed mechanisms that extract particular features from the shared representation without yet transforming them.*

During the afternoon, all jurors work simultaneously on the material they have selected. They do not all perform the same operations. Some jurors have backgrounds that help them combine evidence using sophisticated principles—identifying causal relationships, or drawing careful distinctions that track the case's logic. Others retrieve similar past cases they have heard about, flag inconsistencies, or summarise clusters of facts. Some perform simpler tasks: copying key details, organising information, or highlighting what seems relevant. A few jurors even introduce errors—misremembering a precedent, or drawing hasty inferences. Yet all these contributions, expert and flawed alike, are placed back onto the table at day's end and become material for the next day's deliberation. *This processing phase, where information gets combined and enriched, corresponds to the MLPs.*

As the days progress, the shared body of evidence grows richer and more interconnected. Many lines of thought coexist and influence one another. *A group of jurors who work together across successive days to carry out one procedure is the analogue of a circuit*—distributed across many jurors and many days, with the same juror free to belong to several such groups, just as one attention head or neuron can take part in several circuits.

No single juror sees the complete picture; the emerging understanding arises from how all these parallel contributions interact through the communal table.

By the final day the jury must deliver a collective verdict. That decision emerges from the cumulative interplay of many parallel contributions—some expert, some heuristic, some nearly accidental. Perhaps some jurors' combined reasoning traces the path an expert would have followed in isolation. Yet the final judgment cannot be separated from the surrounding tangle of simpler or less principled inferences layered throughout. *The verdict corresponds to the next-token prediction.*

A summary can be found in the following table.

Jury room element	LLM component
Communal table	Residual stream
A note on the table	A feature
Each day / each jury group	One layer
New jury each day	No memory between layers (except whatever is in the residual stream)
Morning (selecting evidence)	Attention heads
Afternoon (processing evidence, recalling information)	MLPs
A juror group running one procedure across days	A circuit
Jurors' document-selection rules	Attention patterns
Final verdict	Model output (next predicted token)

Table 1. Summary of the jury room analogy.

Two things are worth keeping in mind so the analogy does not suggest the wrong picture. First, a transformer module is far more limited in what it can do per layer than a juror is in an afternoon. A juror might string together many combinations, comparisons, and inferences in sequence; a single attention head or MLP does only about one such step, with perhaps a few more in parallel, never a long sequence. Second, just reading the analogy, one might come away thinking the jurors could not be very coordinated, given how messy the procedure seems. But LLMs have been trained this way from the start and are very good at it, so for the analogy's purposes it is best to imagine the jurors as having worked together for a very long time, as highly coordinated, and as a result a great verdict-making machine.

With the analogy in place, a range of mechanistic-interpretability findings about LLMs becomes intuitive.

Each juror takes only a small subset of what is on the table; correspondingly, attention heads and MLPs attend to only some of the features represented in the residual stream (Elhage et al. 2021). Because so many jurors are at work, items end up on the table in multiple copies, and jurors gravitate toward whatever interests them, whether it appears once or many times. This is not a flaw: the more often a feature is duplicated, the stronger the signal of its importance, and the more likely some attentive juror will pick it up. The same pattern is found in LLMs, where duplicate representations of the same feature populate the residual stream and serve as an implicit importance signal (Lindsey et al. 2025). The preeclampsia diagnosis of §3.1, for instance, was carried not by one feature but by several distinct ones, lying along different directions and firing at different layers.

The redundancy goes beyond duplicates of the same piece of evidence. It also operates at the level of jurors themselves: different groups of jurors arrive at the same conclusion through different routes and submit it in parallel. This is the kind of redundancy behind the *hydra effect*, in which ablating one circuit does not hurt performance much because other circuits compensate for the loss (McGrath et al. 2023). It is also what Chen et al. (2026) observe across many tasks—multiple structurally distinct, low-overlap circuits, each implementing the task on its own—a finding we return to below.

Groups of jurors converging on the same answer might look wasteful, but the redundancy is sometimes doing real work. Their independent contributions can constructively interfere, reinforcing one another where they agree and cancelling out where they do not, yielding a more reliable verdict than any one would produce alone. The modular-addition circuit of §3.2 was exactly this: Fourier circuits at several frequencies reinforced the right answer and cancelled the wrong ones (Nanda et al. 2023).

Different groups of jurors work in very different ways. Valentino et al. (2025) provide a clean illustration: on syllogistic-reasoning tasks, one group of jurors applies the formal logical scheme while another reasons from the content of the premises, leaning on what is empirically plausible rather than on what logically follows. Othello-GPT, a transformer trained to predict legal Othello moves, supplies another: some of its internal circuits represent and use the underlying board state—a principled grasp of the game—while others merely upsample statistically likely next moves based on how many turns have passed, a move-counting heuristic (Nanda et al. 2023). When groups disagree—as in Valentino's syllogisms—one group can end up shouting louder than the other.

Some groups of jurors play more specialised roles in managing the table itself. Some act as amplifiers, copying or rewriting key items in larger letters so they do not get buried under the growing pile. Others act as cleaners, crossing out or covering up items so that later jurors do not pick them up. Dedicated MLPs and attention heads in LLMs do exactly this—repeatedly amplifying a feature so that it stays legible across layers, or suppressing a feature to the point of disabling downstream circuits that would otherwise pick it up (Ameisen et al. 2025).

The table has limited surface area, so jurors constantly write on top of existing evidence using different coloured pens, and read what they need by focusing on the relevant colour (though occasionally the overlapping notes interfere). The residual stream is similarly constrained: it has fixed dimensionality, and networks pack more features into it than there are dimensions by storing them as overlapping vectors—a phenomenon known as superposition (Elhage et al. 2022).

Taken together, these findings deliver the analogy's central lesson. From the outside, it becomes nearly impossible to determine whether the verdict reflects genuine understanding or merely the emergent result of many parallel mechanisms—some sophisticated, some superficial—reinforcing one another through repeated interaction. Much like a transformer's output, the decision is irreducibly the product of this layered, distributed process.

This many-voiced character is what we call *polyphony*. The term is borrowed from Bakhtin's reading of Dostoevsky, where a novel holds many independent voices, none subordinated to a single authorial vision, and the whole emerges from their interplay. The jurors are like this: trained to coordinate, yet none of them the sole author of the verdict. We prefer "polyphonic" to two nearby labels. The phenomenon is not *distributed* in the sense of distributed cognition (Hutchins 1995), which spreads a task outward across many agents and artefacts, whereas polyphony is inward—many voices within what is, from the outside, a single agent. Nor is it *fragmented* or *fractured* (Freeborn 2026), terms that suggest a unity which has broken down, whereas the jury room is a different kind of unity, one that works as it stands and need not be repaired.

With polyphony in view, the task that remains is to draw out what it implies for the concept of understanding, and to adapt that concept to fit LLMs, and deep-learning models more generally.

4.2 Taking polyphony seriously

Polyphony unsettles a cluster of inferences that ordinarily take us from evidence to trust. Begin with the one §3 itself might seem to invite: that locating the right structure inside a model settles whether to trust its output. A puzzle from Vafa et al. (2024) brings out how polyphony complicates even this.

Vafa et al. train a transformer on millions of turn-by-turn New York taxi routes, with the sole task of predicting the next turn, and it predicts a valid turn 96 to 99 per cent of the time. To test whether it has a coherent map of the city, they reconstruct the map implied by its behaviour, and the result is wildly incoherent: streets in impossible configurations, and turns no real city would permit. They conclude that the model has no coherent internal map, and that a probe which seems to recover the current intersection—with an accuracy between 0.91 and 0.99—should not be trusted to show otherwise.

But the same evidence can be read the other way. Reconstructing a single map from behaviour assumes that one map lies behind the behaviour. The behaviour of a polyphonic system, however, is the joint output of many procedures, so a map read off from it blends them together and can look incoherent even when a clean map is present but not in control—outvoted by messier circuits, much as the syllogism circuit was outvoted in Valentino et al. On this reading it is the behaviour-reconstruction that misleads, and the probe's success that should be taken seriously: a largely clean map is there, but it does not govern the routes (Figure 3). Which reading is right, for this particular model, is not our concern. We use the case to illustrate what polyphony does to the reading of evidence.

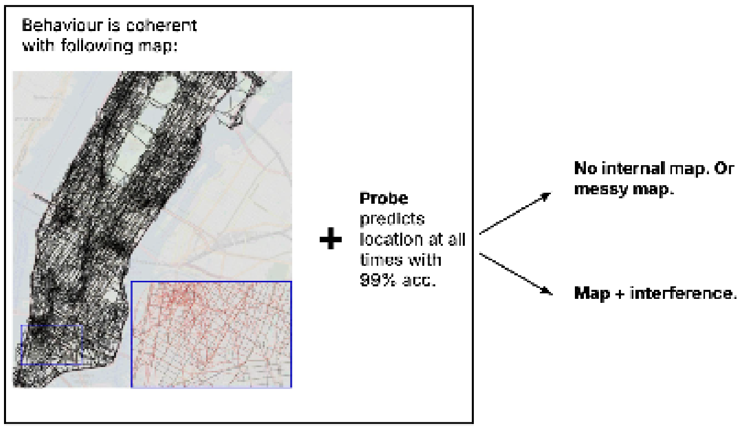

Figure 3. Two readings of Vafa et al.'s (2024) taxi model. The second reading takes polyphony seriously.

The taxi case yields two lessons. The first is that a present circuit is not enough. What the model lacks, on our reading, is not the right structure but the right structure *in control*; the deficit is one of elicitation, of bringing the correct mechanism to the forefront, and as long as behaviour is governed by the messier circuits its routes are not to be trusted, even though a good map exists within. The second is that the converse inference is no safer. Reading a model's competence off the map its behaviour implies, as Vafa et al. do, is not warranted once polyphony is taken seriously: because behaviour pools the contributions of many circuits, an incoherent behaviour-map does not establish that no clean structure is present. Neither behaviour nor internal structure, taken on its own, settles the matter; what trust ultimately calls for is evidence that the right structure is present *and* that it governs the behaviour.

Generalising from the taxi case, a procedure that is present in a model can fail to govern an output in two ways. The taxi model exemplifies the first, *interference*, where the procedure runs but is overpowered by competing ones. The same overpowering appears in the Valentino et al. (2025) syllogism case: the syllogism circuit fires, but the content-sensitive circuits dominate it. Rai et al. (2025) sharpen the pattern with a methodological lever: on balanced-parentheses and arithmetic tasks, models err when faulty mechanisms overshadow sound ones, and steering that amplifies the reliable mechanism restores accuracy from near zero to near 100 per cent—direct evidence that the sound mechanism was there, running, but outvoted. Yu, Merullo, and Pavlick (2023) document the same competition in factual recall: GPT-2 has memory-promoting and context-promoting attention heads, and on prompts with in-context contradictions ("The capital of Poland is London → ?") the answer depends on the relative strength of the two populations.

The second way is *dormancy*, where the procedure does not run at all. It can fail to fire because the case falls outside the range the procedure was built for, with no general method beneath it to fall back on: Nikankin et al. (2025) find that arithmetic in Llama-3 is a bag of heuristic neurons, each firing on narrow operand ranges, so that an out-of-range input matches no member of the bag and the model has no fallback. It can fail to fire because the procedure one identified was specific to the data on which it was discovered, so that a genuinely new case calls up a different group, or none; circuit-discovery work increasingly finds dataset-specific, mixed-mechanism circuits rather than one general task circuit (Rai et al. 2026; Chen et al. 2026), with robust exceptions such as the addition mechanism presented in §3.2 (Feucht et al. 2026). And dormancy can also be induced: the framing of an input can suppress the very procedure that would otherwise run, as when an adversarial suffix lowers a model's harmfulness-feature activation and its refusal lapses (Arditi et al. 2024; Ball, Kreuter & Panickssery 2026).

Neither failure, however, leaves us in the dark. Whether a procedure fires on a given input, and whether its contribution dominates the output, are themselves mechanistic facts, open to the same inspection and intervention that located the procedure in the first place. Polyphony thus relocates the question of trust rather than closing it off—from whether the right structure is present to whether it governs the case at hand.

Polyphony breaks a second inference, and an especially consequential one. In a human understander a cluster of abilities travels together: someone who can explain why the planets move can usually also predict where they will be, say what would happen were things otherwise, and apply the idea to a new case (Hills 2016; Woodward 2003). The concept of understanding leans on this, treating any one ability as a sign of the rest. In a polyphonic model the guarantee lapses, because explaining, predicting, and counterfactual reasoning may be carried by different groups of jurors that need not rely on one another. The abilities can then come apart: a model may explain a domain fluently while predicting it poorly, or answer counterfactuals it cannot turn into predictions. A polished explanation tells us nothing, on its own, about the quality of the predictions; each ability has to be assessed in its own right.

A similar fragmentation appears across a domain. A domain like medicine is not one thing but many: it breaks into sub-domains—the disorders of pregnancy among them—and each sub-domain into individual conditions, such as preeclampsia, and each condition into aspects: its causes, its course, its treatment. To understand medicine a model would need the right structure for each part that matters, and in a polyphonic system each is a separate question: the structure for one condition may be present and that for its neighbour absent, and even where it is present it may fail to fire or be overruled. Asking whether a model understands a domain is therefore asking after a patchwork, piece by piece, not after a single mechanism that either is or is not there.

Two fragmentations, then, and together they mean that understanding a domain comes by degrees, along two axes at once. We can still make the everyday claim that a model understands clinical reasoning, say, or arithmetic, by reading it as a *pattern summary* with one axis for each fragmentation: how much of the domain the model covers, procedure by procedure, and which of the bundled abilities it sustains across them. The claim holds to the degree that, along both axes, the right structure is present, reliably fires, and reliably governs. Read this way, it is a condensed warrant for the inferences a practitioner wants to draw: that the model's verdicts on new cases can be deferred to, that its explanations track the underlying causal structure, that its counterfactuals are structure-respecting.

What this licenses is more informative than the old all-or-nothing verdict, since gaps can open along either axis. Rather than ask whether a model understands “physics” or “medicine” as such, a practitioner can say, along the abilities, that it explains Kepler’s laws but cannot use them to predict where a planet will be, having the structure for the one but not the other; and along the domain, that its grasp of preeclampsia, of the kind §3.1 displayed, extends to the other common disorders of pregnancy but thins out on the rarer ones. Where the whole cluster does hold across a domain, the pattern reassembles what polyphony had separated, and the attribution recovers its inherited force; but it need not, for the concept can equally track an understanding confined to some abilities and some cases.

Polyphony turns a single question—whether a model understands—into many. For each capacity and each part of a domain, attributing understanding now means asking whether the right structure is present, whether it fires on the case at hand, and whether its contribution governs the output. The questions divide by capacity and by domain because both fragment: explaining, predicting, and counterfactual reasoning can come apart, and a domain is covered only in patches. How to answer them when a model is too large to inspect in full is the task of §5.

5. Trusting larger models

Our earlier examples were clean: the circuit could be located, its algorithm written down, and the trust it warranted vindicated by intervention. But they were, if not toy models, then at least relatively small ones, and one might object that they tell us little about the frontier models we actually care about. Those systems are vast, and their size yields two reasons for pessimism about our ability to ground trust in them: they are too large and tangled to reverse-engineer in full, and they have more room to memorise much of what they are trained on.

We resist on both fronts. The memorisation default is weakened by scale, not strengthened: recent work on generalisation suggests that as models get bigger, they track *more* rather than less of their target domain’s structure (§5.1). And the inability to open a model fully does not leave us blind: even short of a complete reverse-engineering, a defeasible case for trust can still be made (§5.2).

5.1 Larger models are more structure-sensitive

A first and general reason to be less wary of deep learning is that the standard by which the field already judges its models is closer to a test for understanding than it looks. That standard is generalisation: a model earns trust by performing well on data held back from training, and better still on data drawn a little differently again. Why should this bear on understanding? Because the only way to do well on data one has not seen is to have *compressed* the data one has in a way that *expresses comprehension*. To handle unseen cases, a model must find regularities or structure in the data that carry over to new cases and thereby render them less surprising; and a structure that goes on predicting data it was never shown has, by that very fact, latched onto something real in the domain rather than onto its particular examples (Delétang et al. 2024; Queloz & Beckmann 2026). That is just what understanding is, on our account: a grasp of the structure that makes a domain hang together. The gold standard the field already trusts is therefore, in effect, a test for the very thing we have been calling understanding.

The argument so far applies to any model that generalises. The pessimist's worry, though, is about size. A frontier model has so many parameters that it has room to store rather than compress, and the more it grows the more room it has, so scale should only make memorising more likely. The evidence runs the other way.

What the field has found is that bigger models generalise better, not worse. The textbook picture expected the opposite. It predicted a sweet spot, with too little capacity leaving a model unable to fit the data and too much leading it to overfit—fitting the training cases at the expense of everything else. Yet the largest models sit well past that point and still generalise better than the mid-sized ones. As model size grows, the error falls, then rises around the point where the model can just fit the data, and then falls again—a pattern known as double descent (Belkin et al. 2019; Nakkiran et al. 2021). And the signature of compression grows stronger with size rather than weaker: larger models, once trained, turn out to be more compressible, to be describable with fewer effective parameters, and to produce lower-complexity outputs (Maddox et al. 2020; Goldblum et al. 2024).

Why should the extra size help at all? An early and appealing guess was the lottery-ticket hypothesis: a bigger network holds more sub-networks, as if it held more tickets, so it is likelier that one of them is a winner that training can bring out (Frankle & Carbin 2019). But later work suggests this is not quite how things work (Martinelli, Brea & Gerstner 2026). Three accounts have emerged in its place.

First, Wilson (2025) argues that flexibility and simplicity, far from trading off, grow together with size: as a model grows, the simple, compressible solutions come to occupy an ever greater share of its space of possible solutions, so that training, which has to land somewhere, ever more likely lands on one of them (Figure 4, left). The model could memorise; with overwhelming likelihood, it turns out not to. On this reading, the textbook's mistake was its measure of complexity: it judged a model by the size of the space it could search, when what matters is the simplicity of the solution it settles on.

Second, larger models are less likely to get stuck in a dip they cannot climb out of. Training does not survey the space of solutions from above; it starts somewhere and walks downhill—towards lower-loss configurations—until it can go no lower. A small model tends to get caught in a suboptimal spot along the way, because every direction around it leads back up. Martinelli, Brea, and Gerstner (2026), drawing together a longer line of results (Fukumizu & Amari 2000; Simsek et al. 2021), describe what extra width does to this. The new directions it opens can serve as escape dimensions: a dip that traps a narrow model becomes, in a wider one, a place the search can still roll out of, because one of the added directions leads downhill (Figure 4, middle). As the model grows, these traps turn into mere way-stations and grow rare beside the good solutions, so the search is far likelier to settle in a good one. The extra parameters that looked like room for memorising do different work: they make the loss landscape more navigable.

Third, a model's size affects which procedures can take shape inside it as it learns, which persist, and which are overwritten. Huang et al. (2026) study a model learning a few common tasks alongside many rare or intricate ones—the shape of the data language models are in fact trained on (Figure 4, right). A small model spends what capacity it has keeping up with the common tasks, and any rare, structured procedure that begins to form is overwritten by the next wave of common updates before it can consolidate; rare learning becomes, in their words, an update-and-forget loop—one that never closes, since, fed unlimited data, the small model still never learns the rare task. Make the model larger and two things follow. First, with capacity to spare, the frequent tasks are fully mastered and their gradients fall quiet, so they no longer overwrite anything else. Second, freed of that overwriting, the trace a rare task leaves on one encounter is retained rather than wiped, accumulating across sparse appearances until it consolidates into a procedure that generalises.

It helps to picture the result as two jury rooms, a small one and a large one, each learning over many cases to deliver good verdicts. The small room has too few jurors to go round. Everyone is tied up keeping pace with the steady stream of routine cases, which they handle perfectly well, and no one can be spared, so whatever expertise starts to form on the rare, hard cases is trampled by the next batch before it sets. The large room has jurors to spare: once enough of them cover the routine work, the rest are free to work slowly on the rare cases, turning scattered notes into real expertise that survives from one such case to the next until it hardens.

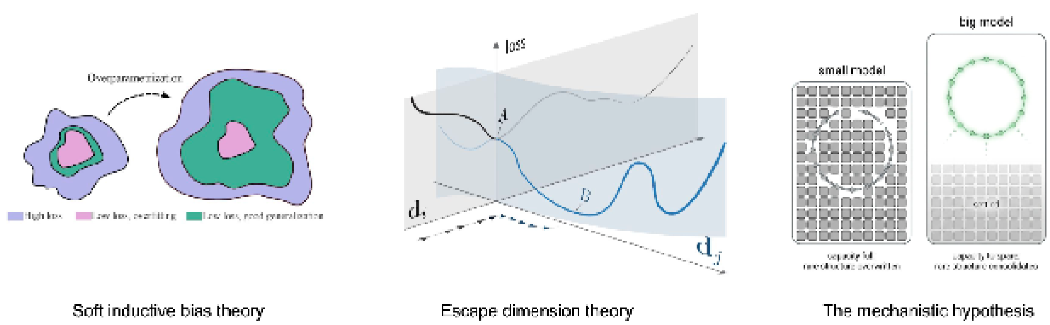

Figure 4. Three accounts of why larger models generalise better. *Left, soft inductive bias.* The space of solutions a model could settle into, each point a setting of its weights. Most settings fit the data poorly (purple); a smaller region fits it well, and there the solutions that merely overfit (pink) sit alongside those that also generalise (green). Overparametrisation enlarges the space and, with it, the share taken up by generalising solutions (Wilson 2025). *Middle, escape dimensions.* Loss along the model's weight directions. Seen along one direction, d_i, the search settles at A, a low point with no downhill exit. Widening the model adds another direction, d_j, along which A is no longer a dead end: the search can carry on down, following B to a better solution. As width grows, such traps increasingly turn into routes through to better solutions (Martinelli, Brea & Gerstner 2026). *Right, the mechanistic hypothesis.* In a small model, capacity is taken up by the frequent cases, and a rare structured procedure is overwritten before it can form. In a larger model there is more capacity to spare, so the rare procedure can take shape and is retained across its sparse appearances until it consolidates (our visual summary; Huang et al. 2026).

One qualification matters. All of this holds in the regime where a model is far smaller than the data it learns from, and that is the regime the large models are in: their training corpora run into the trillions of words, far more than even hundreds of billions of parameters could store, so fitting the data means compressing it rather than keeping it. A model as large as its data could simply store it, and there the pessimist's fear would be right; but that is the reverse of the case at hand.

Where the data so far outruns the model, size is a reason to expect more structure, not less.

One concession is owed to the pessimist: scale buys more verbatim memorisation too (Tirumala et al. 2022); a larger model is simply more polyphonic, carrying more stored cases and more structure-tracking procedures side by side. But even for a model too large to reverse-engineer in full, the default expectation should be that structure is in there.

None of this licenses blind trust. It shifts the prior and no more—a claim about the base rate, not a verdict on any particular model or case. With LLMs the argument is harder to run cleanly, since they are trained on so much that genuinely held-out data is hard to come by; but the question it turns on is unchanged: how far the procedures a model has formed carry to data it was not trained on.

5.2 From ideal to actionable indicators

Trusting a model comes down to whether the right structure is present and does the work on the case at hand, and how conclusively this can be shown depends on the model. It helps to mark two tiers of evidence. The ideal tier is a full verification, and it consists in four converging things. *Circuit identification:* we can point to the specific attention heads, MLPs, or feature directions that implement the procedure. *Algorithmic characterisation:* we can write down the algorithm that circuit computes, in terms an outside observer could run on paper. *Causal verification:* we can intervene on the circuit—ablating it, patching in other values, steering its activations—and the output changes as predicted. *Domain generalisation:* the circuit handles inputs across the domain, including ones unseen in training, as the characterisation says it should. Modular addition is the canonical instance, all four met, as we saw: a Fourier circuit identified in particular heads and MLPs, its algorithm written down, its causal role confirmed by intervention, and its reach established across every operand pair. When the four converge, the ascription is not merely well-supported but vindicated: we have written down what the model is doing, shown that this is what it is doing, and shown that doing this is what handles the cases.

Frontier models defeat full verification. Their circuits are too large, too entangled, and too multi-functional to identify in whole, and an intervention on one part can be read only against assumptions about the rest. The ideal tier keeps its value even so, since it shows that the conditions for ascribing understanding can actually be met in particular systems, and supplies the verified cases against which weaker evidence is checked; but for the frontier we need a coarser, second tier. Some philosophers of AI conclude that, without full verification, there is simply no telling whether such a model has grasped the structure or merely struck lucky.

That overstates the loss, and the correction is to invert it. It is tempting to say we cannot really know what goes on inside these models, but the thought runs backwards. We know more about what a small transformer does in adding two numbers than about what a mathematician does in the same sum; and even frontier models, hard as they are, are not closed to us. The system cards now published with large models lean on interpretability: they locate internal features—an evaluation-awareness feature among them—and steer those features to watch behaviour change, well short of reverse-engineering the network but well beyond anything available for a brain (Anthropic 2025). This is done in the service of safety rather than to certify understanding, but it settles the question of feasibility: the methods reach frontier scale. Reading a model's activity and intervening on it is a real advantage the study of human understanding never had, and the frontier limits leave a great deal of evidence standing. The coarser, actionable tier is built from it.

The actionable evidence comes in two kinds. The first are *mechanistic signatures*: signs that the right features or circuits are present and do causal work, short of any algorithm being written down. A feature that fires on inputs sharing a structural rather than a surface property, composes with others in domain-respecting ways, and sits in an attribution graph routing through interpretable steps is evidence of just this kind. The preeclampsia diagnosis was an instance: structured features composing in a domain-respecting way, their role shown by intervention, even though no algorithm was written down as it was for modular addition. The gap is the point. A circuit located but not fully reverse-engineered—evidence that the right procedure fired, before its algorithm can be read off—already counts. And the same tools, shown feasible by the safety work, can be turned on trust-relevant structure: whether a robust circuit such as the reused base-10 addition mechanism of Figure 2 has formed, whether the right features fire, even how many such circuits a model carries. These are indicators of circuits, not reverse-engineerings of them.

The second kind are *behavioural signatures* the mechanistic story predicts. A procedure that has captured a domain's structure should let the model handle in-domain cases it never saw, answer counterfactuals in structure-respecting ways, hold up under irrelevant surface variation, and stay consistent across cases the structure groups together. The first of these weighs most, and here the compression point pays off: succeeding on cases one has not seen is possible only by having compressed the structure that generates them, so strong performance on genuinely new data is the output of structure-tracking itself—the nearest a behavioural test comes to reading it directly. The difficulty is that a model trained on almost everything leaves little data that is surely new; the answer is to make some—a test set built after the model was trained, or designed to fall outside it. High accuracy there is a strong, if still defeasible, sign.

Two things keep this from being a collection of plausible-looking heuristics, and both bear hardest on the large models that need them most. The indicators are *calibrated*: in the smaller models we can fully verify, the features that fire are the very ones the verified circuit uses, and broad generalisation is just what such circuits produce, while its absence marks competence still tied to the training data. So when the same signature turns up in a model we cannot open, it earns the same reading—on a correlation that is empirical and will be revised as interpretability matures. And the reading does not begin from neutral. A model that generalises well is already more likely to be tracking structure than storing cases, the more so the larger it is, as we argued; so on a frontier model the actionable indicators start from a prior that favours structure and raise it further.

Even so, no single indicator is decisive. Structured features can be present without driving the output, as the taxi model warned, and success on unfamiliar cases can come from a fortunate alignment of heuristics rather than from a procedure that has grasped the structure. The indicators work by accumulation: each is partial and defeasible, but they converge across mechanistic and behavioural lines, across several probes, and across cases of varying difficulty, and the convergence is what builds the case. This is the ordinary shape of evidence for something that cannot be seen directly and has to be triangulated from imperfect signs.

That is enough to meet the pessimist. Trust at the frontier is reachable without full reverse-engineering: a freshly built dataset handled accurately, or a check that the right procedure fired, is a genuine indicator. What it yields is not the settled verdict of the ideal tier but a graded and revisable warrant—which is the most that trusting a system of this size could ever rest on.

Drawing the three sections together: a model's competence is grounded in inspectable structure, and where that structure is of the right kind and does the work, it grounds trust. That is what understanding comes to here. Two things complicate the step from structure to trust. The structure is polyphonic, so locating a procedure does not guarantee it governs the output, and a model's capacities come apart, so understanding has to be credited capacity by capacity and case by case rather than read off a single performance. And the models we most need to judge are too large to verify in full, so trust rests on a default that scale tilts towards structure rather than on a complete reverse-engineering.

Whether a given model understands a given thing then comes down to a handful of questions:

- *Probe-decodable.* Can the domain's structure be read off the model's internal states?
- *Causally used.* Does intervening on that structure move the output as the structure would predict?
- *Circuit-characterisable.* Can the mechanism be isolated as a circuit running a specifiable algorithm?
- *Perturbation-robust.* Does it hold under structure-preserving changes and break under structure-violating ones?
- *Elicitation-robust.* Does the procedure reliably fire, and carry, across the cases at issue?
- *Scope.* For which capacities (explaining, predicting, counterfactual reasoning), and across how much of the domain, do the answers hold?

How conclusively these can be answered falls into two tiers:

- *Ideal.* On a small system open to reverse-engineering, each question is settled outright and the ascription is vindicated, as with modular addition.
- *Actionable.* On a frontier model, full verification is out of reach, so the questions are met by coarser indicators: mechanistic signatures (a feature located, its causal role shown) and behavioural ones (structure-respecting generalisation, above all on freshly held-out data). Calibrated against the verifiable cases and taken together, these accrue defeasible but genuine warrant.

A claim that a model understands a whole domain is then a pattern summary over these answers.

6. From polyphony to unified understanding

The picture so far has set a unified human understander against a polyphonic model, its competence spread across many mechanisms with no single locus. At this stage two objections might be pressed against it. One comes from the human side: cognitive science has long described the brain itself as a crowd of parallel, competing processes rather than a single reasoner (Kahneman 2011), so the unified understander looks less like a fact about us than an idealisation. A second comes from the other direction: agentic systems such as Claude Code now behave with such consistency across long tasks that dwelling on their sprawling, polyphonic interior can seem like over-reading, when what they do is just what an understander would do.

We think the two can be answered together. The inherited concept presupposes a unified, coherent understander; inside, both systems are polyphonic, their competence spread across many voices rather than gathered in one. With so many voices at work one might expect incoherence, and that is just where elicitation earns its place: the coherence the concept presupposes is not intrinsic to either system but achieved, when it is, by scaffolding. Given the right kind of elicitation, a polyphonic interior can be brought into a form that behaves as a unified understander at the level where the concept does its work. We show how this happens for humans—through pen and paper, notation, and a surrounding community—and how it may be coming to happen for AI models—through chain-of-thought and the agentic loop—drawing the parallels and marking the distinctions as we go.

Start with the human case, where the concept of understanding is at home. Grant the human-side objection at once: the brain may well be polyphonic, as Kahneman says, a crowd of competing processes with no single locus. It does not follow that the understanding epistemologists prize sits at that level. There is good reason to think it may lie elsewhere—with the knower's tools and community, not in the brain alone.

We would suggest this holds even of the accounts that look most individualist. The understanding Hills, Kvanvig, and Grimm analyse is typically a scientist's, and a scientist is never just a brain: she works with notation and instruments, within a community whose standards her grasp must meet, so the understander even these accounts have in view is arguably already an equipped and embedded one. Others make the dependence explicit. Boge, Schuster, and Stoll (2026) distinguish a *scientist's* understanding—an individual's grasp, which may be tacit and inarticulate—from *scientific* understanding, which a phenomenon enjoys only once there is an explanation communicable to a community. And Queloz and Beckmann (2026) make sociability itself a shaping force: the demand to make understanding communicable, the pressure of an internalised community, presses the individual towards a more compressed and elegant grasp than a solitary brain would reach. The elegant end of understanding, on all these views, is achieved with the social and the external, not by the bare brain alone.

Consider my own understanding of planetary motion. What I retain of Kepler's laws, from a class long ago, may lie in my head as a sprawl of half-connected fragments—could one look in, it might present as just what Freeborn (2026) calls a fractured understanding in deep nets. When I have to work out where a planet will be some days from now, what draws the fragments into something elegant is pen and paper. I try a relation, see it does not fit, backtrack, cross things out, drop the steps that come out inconsistent, and as I write they settle into a clean procedure that solves the problem. The elegance is achieved on the page, not read off the head. It is me-plus-pen-and-paper that is the subject of the understanding. And the same work does more: having reasoned the prediction through, I am better placed to explain the laws too, so the scaffold that drew my grasp into coherent form has also begun to knit its capacities together.

The model's side runs the same way, one level down. Its competence is spread across many parallel circuits, no one of them the whole, and chain-of-thought is the medium in which the right ones get selected: the model floats a line, checks it, backtracks when it fails, and settles on the procedure that works. A recent result shows how much of this is selection rather than fresh capability. Venhoff et al. (2025) first read off, from a fully trained "thinking" model, a vocabulary of reasoning moves (planning, checking, backtracking, computing), each a recognisable pattern in its activations, and confirm that the same patterns are already present in the base model that never had reasoning training. They then build a hybrid: the base model runs as usual, but at each step a classifier judges which move should come next and nudges the base model's activations in that direction—with no retraining, and on only about 12% of tokens. That alone recovers most of the gap to the full thinking model: up to 82% on GSM8K and 91% on MATH500. The base model already knew how to make each move; what it lacked, and what the steering supplies from outside, is the sense of when to make which.

This is the model's version of working a problem out on paper. What I retained of Kepler's laws was all there, but unsequenced; the page gave me somewhere to try a move, see it fail, cross it out, and put it in order. Chain-of-thought is where the model's moves get tried, checked, and sequenced in the same way. The moves, in both cases, were already inside; the scaffold supplies their sequence, not the capacity.

Pen and paper for us, chain-of-thought and the agentic loop for the model, do the same kind of work, which we can call *elicitation*: each draws a coherent procedure out of a polyphonic interior and holds the many voices to it. Agentic settings push the model's elicitation furthest, since external feedback—a test that passes, or a compiler that complains—selects among its circuits in a way plain chain-of-thought cannot. This answers, one level up, the question polyphony left us with: whether the right procedure fires, and whether it governs the output. Elicitation secures both across a stretch of work; in neither system is the coherence a property of the bare substrate, and it is elicitation that makes the right voice sound and carry.

None of this requires the parallel to be exact. The two kinds of elicitation may work quite differently, and may foster understanding of different kinds; we should not assume that what holds the model's voices together is what holds ours. What the parallel suggests is rather a possibility. For us, the concept of understanding already abstracts from the polyphonic brain and works at the level of behaviour: we do not look inside the head to credit it. It may be that, as elicitation in models improves, the concept comes to work the same way for them—abstracting from the polyphonic interior and resting on behaviour, at the level where it has always done its work. We are not there yet. These models still behave in strange ways, which tells against crediting them with understanding at the behavioural level; and in many cases of interest the right procedure is not reliably elicited, as the taxi model showed (Vafa et al. 2024). For now, looking inside still matters.

Conclusion

We began from a question that practitioners cannot evade and theorists have left stalemated: whether, and in what sense, an AI model understands what it is doing. Our answer has been to refuse both the deflationist's dismissal and the anthropomorphiser's credulity, and to treat the concept of understanding as something to be re-engineered for the AI case. The cognitive register is indispensable, because any vocabulary fine-grained enough to separate trustworthy outputs from lucky ones reintroduces the inferential core of understanding; but the inherited concept must be adapted—its tie to structure-tracking competence preserved, its human-specific commitments severed, and new links forged to the internal structure that interpretability can expose.

What results is not a verdict but a framework. Because a model's competence is polyphonic, and at the frontier only partially open to inspection, attributions of machine understanding are task-indexed, graded, and defeasible, and the trust they license is something a model's inspectable organisation earns rather than something its fluency presumes. The conditions of trust are accordingly plural, and answerable one at a time: whether the right structure is present, whether it governs the case at hand, and how far it carries across a domain. Stated this way, the question of machine understanding becomes tractable—a matter of asking, condition by condition, what an AI must grasp before we rely on it, and how far it has done so.